FORMAL SCHEMES ARE SHEAVES FOR THE FPQC TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The last sentence of the proof is the author's 'We omit the verification that these morphisms glue to the desired morphism Spec(R) to X.' It is reproduced verbatim and recorded as an obligation of the card; everything up to the glueing, including the existence of the local morphisms, is proved in the span.

Editorial note. The statement carries a slogan environment in this chapter; the chapter's own preamble defines slogan as a comment environment, so such material is discarded at build time and is not part of the delivered mathematical text.

Editorial note. The proof refers to the description of morphisms from an affine formal scheme by continuous ring maps; that display is recalled by reference and carried as a navigation stub.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The sheaf condition for the fpqc topology tested on Zariski coverings and single faithfully flat maps. Topologies, Lemma topologies-lemma-sheaf-property-fpqc, invoked at proof lines 654-656. Reduces the sheaf condition to the two cases treated.
- The functor of points of a formal scheme and the sheaf condition for Zariski coverings. Same chapter, Lemma lemma-fully-faithful, invoked in the statement at line 650 and at proof lines 656-658. Fixes the functor and supplies the Zariski half of the reduction.
- Descent of maps of sets along a faithfully flat ring map, and the quotient topology in that situation. Descent, Lemma descent-lemma-equiv-fibre-product at proof lines 664-666, and Morphisms, Lemma morphisms-lemma-fpqc-quotient-topology at proof lines 668-670. Produce the set-theoretic descent and then continuity of the descended map.
- The equalizer of a faithfully flat ring map is the base, and morphisms from an affine formal scheme correspond to continuous ring maps. Descent, Lemma descent-lemma-ff-exact at proof lines 686-688, together with the

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display equation-morphisms-affine-formal-schemes used at proof lines 681-683 and 691-692. Descend the ring map defining the morphism on each chosen open and translate it back into a morphism.

Source reference. Original lemma 02KI: sources/stacks-chow/tags.txt, source lines 3373–3373. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 023M: sources/stacks-chow/tags.txt, source lines 2747–2747. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original equation 0AHZ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 470–470. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AI1: sources/stacks-algebraic/pinned-original-source.tex, source lines 2746–2746. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02JY: sources/stacks-chow/tags.txt, source lines 3353–3353. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 022H: sources/stacks-chow/tags.txt, source lines 2706–2706. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $\mathfrak{X}$ be a formal scheme. The functor of points $h_{\mathfrak{X}}$ (see Lemma 0AI1) satisfies the sheaf condition for fpqc coverings.*

Proof. Topologies, Lemma 022H reduces us to the case of a Zariski covering and a covering $\{\mathrm{Spec}(S) \to \mathrm{Spec}(R)\}$ with $R \to S$ faithfully flat. We observed in the proof of Lemma 0AI1 that $h_{\mathfrak{X}}$ satisfies the sheaf condition for Zariski coverings.

Suppose that $R \to S$ is a faithfully flat ring map. Denote $\pi : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ the corresponding morphism of schemes. It is surjective and flat. Let $f : \mathrm{Spec}(S) \to \mathfrak{X}$ be a morphism such that $f \circ \mathrm{pr}_1 = f \circ \mathrm{pr}_2$ as maps $\mathrm{Spec}(S \otimes_R S) \to \mathfrak{X}$. By Descent, Lemma 02KI we see that as a map on the underlying sets f is of the form $f = g \circ \pi$ for some (set theoretic) map $g : \mathrm{Spec}(R) \to \mathfrak{X}$. By Morphisms, Lemma 02JY and the fact that f is continuous we see that g is continuous.

Pick $y \in \mathrm{Spec}(R)$. Choose $\mathfrak{U} \subset \mathfrak{X}$ an affine formal open subscheme containing $g(y)$. Say $\mathfrak{U} = \mathrm{Spf}(A)$ for some admissible topological ring A. By the above we may choose an $r \in R$ such that $y \in D(r) \subset g^{-1}(\mathfrak{U})$. The restriction of f to $\pi^{-1}(D(r))$ into $\mathfrak{U}$ corresponds to a continuous ring map $A \to S_r$ by (0AHZ). The two induced ring maps $A \to S_r \otimes_{R_r} S_r = (S \otimes_R S)_r$ are equal by assumption on f. Note that $R_r \to S_r$ is faithfully flat. By Descent, Lemma 023M the equalizer of the two

arrows $S_r \to S_r \otimes_{R_r} S_r$ is R_r. We conclude that $A \to S_r$ factors uniquely through a map $A \to R_r$ which is also continuous as it has the same (open) kernel as the map $A \to S_r$. This map in turn gives a morphism $D(r) \to \mathfrak{U}$ by (0AHZ).

What have we proved so far? We have shown that for any $y \in \mathrm{Spec}(R)$ there exists a standard affine open $y \in D(r) \subset \mathrm{Spec}(R)$ such that the morphism $f|_{\pi^{-1}(D(r))} : \pi^{-1}(D(r)) \to \mathfrak{X}$ factors uniquely though some morphism $D(r) \to \mathfrak{X}$. We omit the verification that these morphisms glue to the desired morphism $\mathrm{Spec}(R) \to \mathfrak{X}$. $\square$